

Proposal

Soft Cushions

High-level goal

Our idea is to create an interactive Resolutions Tracker website that helps people stick to their New Year's health goals by letting them log daily data like exercise, calories, water intake, sleep, and habits like brushing teeth or reading. The website will update a dashboard with graphs showing their progress over time, giving them a clear visual of how they're doing.

Description of your goals

The goal of this project is to build a tool that encourages individuals to stay accountable to their health goals by tracking their daily habits in a fun, interactive, and visual way. By allowing users to input their health data, the tracker will give them real-time feedback and insights into their progress through graphs and visualizations. The motivation behind this project stems from our personal interests in promoting healthy habits and the idea that visualizing progress can be a powerful tool for motivation. Many people struggle to maintain their New Year's resolutions, especially health-related ones, and this tracker aims to make that process easier and more engaging. We want to create something that not only helps people track their health metrics but also motivates them to continue improving. We plan to use Quarto for building the site and R for data processing and creating visualizations using tools like ggplot2. Additionally, we'll be adding shinyjs to introduce interactive features, and user data will be stored in a simple database or CSV format for easy updates. The project also aims to make the experience more engaging by integrating user login authentication using OAuth, awarding badges for goal completion, and creating interactive graphs using plotly or highcharter for enhanced user interaction with their progress.

For this project, we created our own dataset to simulate real sample user's health tracking data, stored as Sample_User_Health_Data.csv in the data/ folder. Since health tracking data is highly personal and not readily available in public repositories, we decided to build the dataset ourselves so that it could fully reflect the metrics our tracker is designed to monitor. This dataset serves as a completed guide to illustrate the final version of our product when users enter their own data. To ensure functionality for users to view this sample user data as a reference and enter in their own information, we need to temporarily store the dataset in our Shiny app files, so that does not require a dataset to be permanently stored within this data folder.

```
sample_data <- read.csv("../data/Sample_User_Health_Data.csv")
```

The dataset contains daily health log entries across a comprehensive set of metrics aligned with common New Year's health resolutions. Each row represents one day of logged data for a single user. The variables included are: `date` (the calendar date of the entry in YYYY-MM-DD format), `exercise_minutes` (total minutes of physical activity logged that day), `type_of_exercise` (the category of exercise performed, such as Cardio, Strength, or Flexibility), `intensity_of_exercise` (the self-reported intensity level: Low, Moderate, or High), `steps` (total step count for the day), `calories_eaten` (total caloric intake), `protein_grams`, `carb_grams`, and `fat_grams` (macronutrient breakdown in grams), `water_cups_drunk` (water consumed in cups), `hours_of_sleep` (hours of sleep the previous night), `sleep_quality` (a self-rated sleep quality score on a scale of 1 to 5, where 1 is poor and 5 is excellent), `pages_read` (number of pages read that day), `meditation_minutes` (minutes spent meditating), `fruit_servings` and `vegetable_servings` (daily servings of fruits and vegetables respectively), `weight` (body weight in kilograms), and `muscle_percentage` (estimated muscle mass percentage).

The dataset spans approximately six months of sample entries, from January 1 to June 29, 2025, providing a rich range of data to demonstrate meaningful trends in the dashboard visualizations. It has 180 rows and 18 columns. We chose to create a synthetic dataset rather than use an existing public one because health tracking data is highly personal, and a purpose-built sample allows us to demonstrate all the features of our tracker, including streaks, badge milestones, and multi-metric progress trends, in a controlled and illustrative way.

Weekly plan of attack

Week 1: Project Setup and Initial Planning

Xinyi: Set up Quarto project structure and initialize GitHub repository. Amanda: Plan the website layout, including where to input data (exercise, calories, water intake, etc.) and how the dashboard will display progress. Shence: Design basic input forms (using Shiny) for daily health metrics. Amanda: Research OAuth for user authentication and start planning how to integrate it into the website.

Week 2: Data Structure and Basic Interactivity

Xinyi: Create a system to store user data (CSV or SQLite database) and ensure the data can be dynamically updated. Shence: Start implementing basic ggplot2 visualizations to display user progress (e.g., line charts for tracking exercise, bar charts for calories). Shence: Implement basic shinyjs features for input validation (e.g., preventing incorrect data entry). Amanda: Begin integrating OAuth for user login, testing authentication flow.

Week 3: Advanced Features and Visualizations

Amanda: Finish integrating OAuth for secure login, enabling users to save and retrieve their data. Xinyi: Create advanced plotly or highcharter visualizations for more interactive graphs (e.g., allowing users to zoom in on trends). Xinyi: Add features for setting and tracking health goals (e.g., calorie targets, sleep hours). Shence: Implement shinyjs to trigger badges or milestones when users reach their goals (e.g., badges for 7 consecutive days of exercise).

Week 4: Data Storage Enhancements and Gamification

Shence: Develop and integrate the badge system for rewarding users who reach health milestones (e.g., workout streaks, reaching calorie targets). Amanda: Implement data export functionality so users can download their health logs. Xinyi: Refine UI for clarity, adding tooltips and instructions to guide users through data entry.

Week 5: Testing, Debugging, and Final Preparations

Xinyi: Conduct comprehensive testing of the app, focusing on edge cases (e.g., missing data, incorrect formats) and ensuring smooth functionality. Amanda: Finalize all features and ensure everything works as expected, with special attention to data entry, visualization, and user authentication. Shence: Clean up the code, ensuring it's modular, well-commented, and easy to follow. TEAM: Write the final report explaining your design choices, implementation process, and lessons learned. TEAM: Prepare for the oral presentation, focusing on the project's impact, how it meets user needs, and the technologies used.

Final repository organization

The repository is organized to keep the project clear and easy to follow. The root folder contains `README.md` to explain the project, `proj-02.Rproj` for RStudio setup, `DESCRIPTION` for package metadata, `renv.lock` and the `renv/` folder to ensure the R environment can be fully reproduced across machines, and `air.toml` for code formatting configuration.

The repository has three subfolders: `app/`, `data/`, and `reports/`. The `app/` folder contains the Shiny application. `app.R` runs the app and controls the interface and server logic, while `authentication.R` handles user login via OAuth. A `README.md` is included to document what this folder is for and how to run the app. The `data/` folder stores the dataset used in the project. Currently it holds `Sample_User_Health_Data.csv`, a manually created dataset of 180 days of simulated health-tracking data. The `README.md` in this folder acts as a codebook, describing every column, its type, and its meaning. The `reports/` folder contains all Quarto documents: `proposal.qmd` and its rendered output `proposal.pdf`, as well as `finalreport.qmd` for the final report. A `README.md` explains the folder and how to render the documents.

Overall, this structure separates the application code, the data, and the written reports into distinct folders. This makes the project easier to navigate, easier to read and easier for collaboration